Triology Question (4)

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CMR

① You are a book collector aiming to sell A books.
Thickness : B books [Array B]
To enhance appeal, protective film must exist at least one subarray size C such that books with protective greater than books w/o cover.

5
2 3 5 1 4 } → output 2.
3

② You are required to answer q queries, where in each query, you will be provided values l & r.
For each query.
P (code) will fail to print found with any value i.
Express P → int.
P/Q and gcd (P, Q) = 1
P * Q^ -1 modulo 10^9 + 7
where Q^ -1 ⇒ multiplicative inverse ⇒ Q modulo 10^9+7.

eg: A = [[2, 9]] → [500000004]

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(2) Given the pseudocode below
find a target num i bw l and r

You r req to answer q queries, where in each query you will be provided with values l and r. For each query, determine :-

What prob that the above code will fail to print (" Found ") when any value i (where l<=i<=r) is chosen ? Express this probability as an integer where fraction is P/Q and gcd(P, Q) = 1. You should compute P * Q^-1 modulo 10^9 + 7, where Q^-1 denotes the mul inverse of Q modulo 10^9 + 7.

(3) An array is considered valid if there is no subarray longer than B where all elements are identical. Aditionally, the array can only contain integers from range [1, C]. Determine the number of valid array length A. Since the result can be very large, return it modulo (10^9 + 7).

(4) given arr consisting of N integers. You are also given q queries. Each consisting of three int x, y, z.

For each Query :-

Find the largest contiguous subarray B starting from index x whose y^th bit is set.

(3)

arr → N ints.

q → queries. (x, y, z)

for each query →
largest contiguous subarray B.
index x whose yth bit is set.
each of its elements B_j with $B_j \wedge z$
print total ~~queries~~ no. of updates
after performing q queries.

eg:
[2, 4, 3, 5, 4]
[[3, 1, 4], [2, 3, 7]]

4

(1) An array is considered valid
if no subarray longer than B. where all elements identical.

A → array length will be A.
B → B is max identical.
C → range of ele (1, C) → in array
return integer where no. of ~~identical~~ valid
array found.

eg: 3
1
3

valid arrays:
[1,2,1], [1,2,3], [1,3,1]
[1,3,2], [2,1,3], [2,1,2], [2,3,1],
[2,3,2], [3,1,2], [3,1,3], [3,2,1],
[3,2,3]

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update each elements Bj with Bj ^ z

where ^ denotes bitwise XOR

Your task is to print the total numbers of updates performed after q queries.